

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**



Lecture No.- 04

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Topics to be Covered

Topic



- 1:- 8×1 mux by using 2×1 mux
- 2:- 8×1 mux by using 4×1 mux – special case
(Enabling System)
- 3:- 16×1 mux using 4×1 demux
- 4:- 16×1 mux using 2×1 demux
- 5:- Demultiplexer- $1 \times 2, 1 \times 4, 1 \times 8, 1 \times 16$ Demux



8x1 Multiplexer

In this section, let us implement 8x1 Multiplexer using 4x1 Multiplexers and 2x1 Multiplexer. We know that 4x1 Multiplexer has 4 data inputs, 2 selection lines and one output. Whereas, 8x1 Multiplexer has 8 data inputs, 3 selection lines and one output.

So, we require two **4x1 Multiplexers** in first stage in order to get the 8 data inputs. Since, each 4x1 Multiplexer produces one output, we require a **2x1 Multiplexer** in second stage by considering the outputs of first stage as inputs and to produce the final output.



Topic : 8x1 Multiplexer



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Let the 8×1 Multiplexer has eight data inputs I_7 to I_0 , three selection lines s_2 , s_1 & s_0 and one output Y . The **Truth table** of 8×1 Multiplexer is shown below.

Selection Inputs			Output
s_2	s_1	s_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

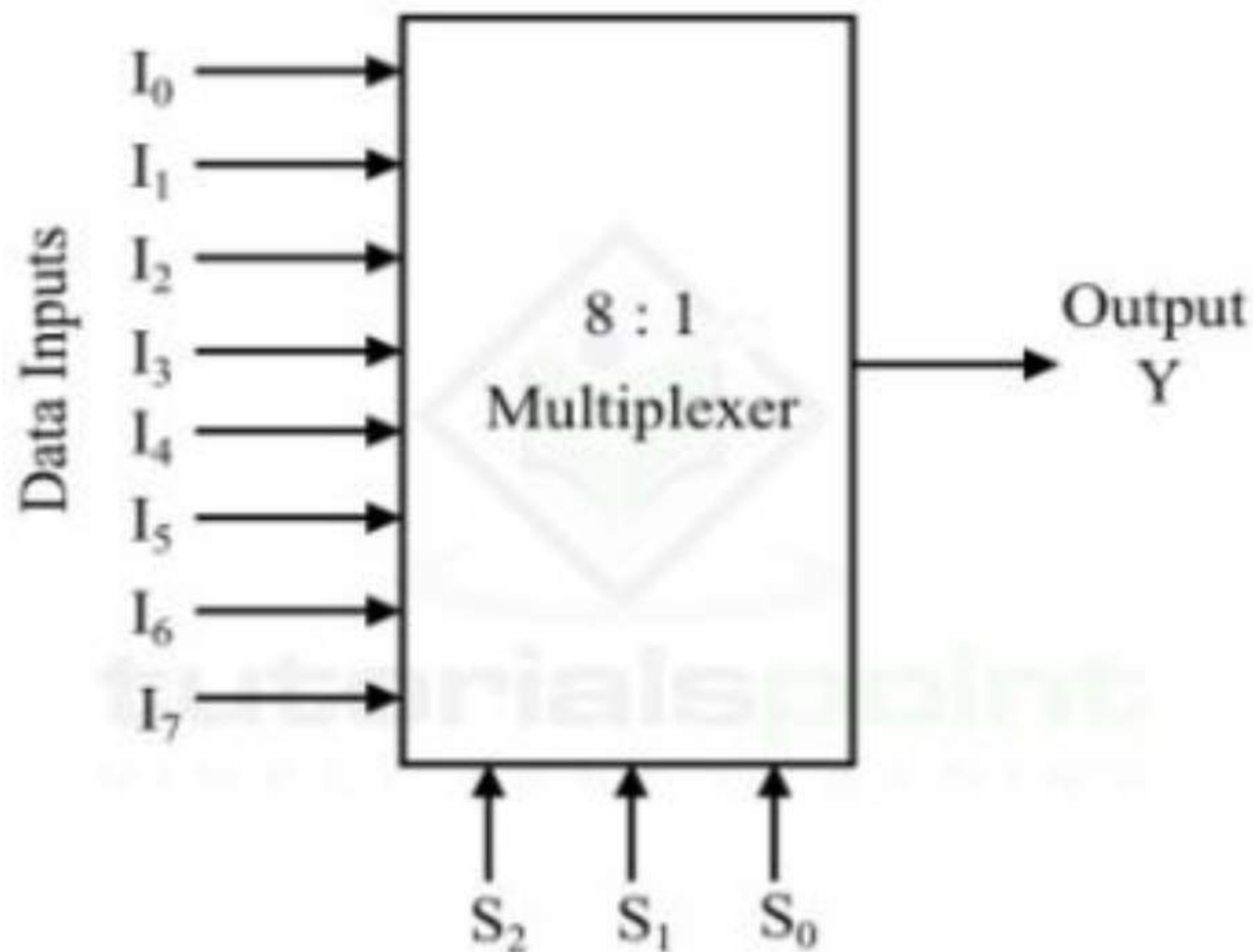


Topic : 8x1 Multiplexer



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We can implement 8×1 Multiplexer using lower order Multiplexers easily by considering the above Truth table. The **block diagram** of 8×1 Multiplexer is shown in the following figure.

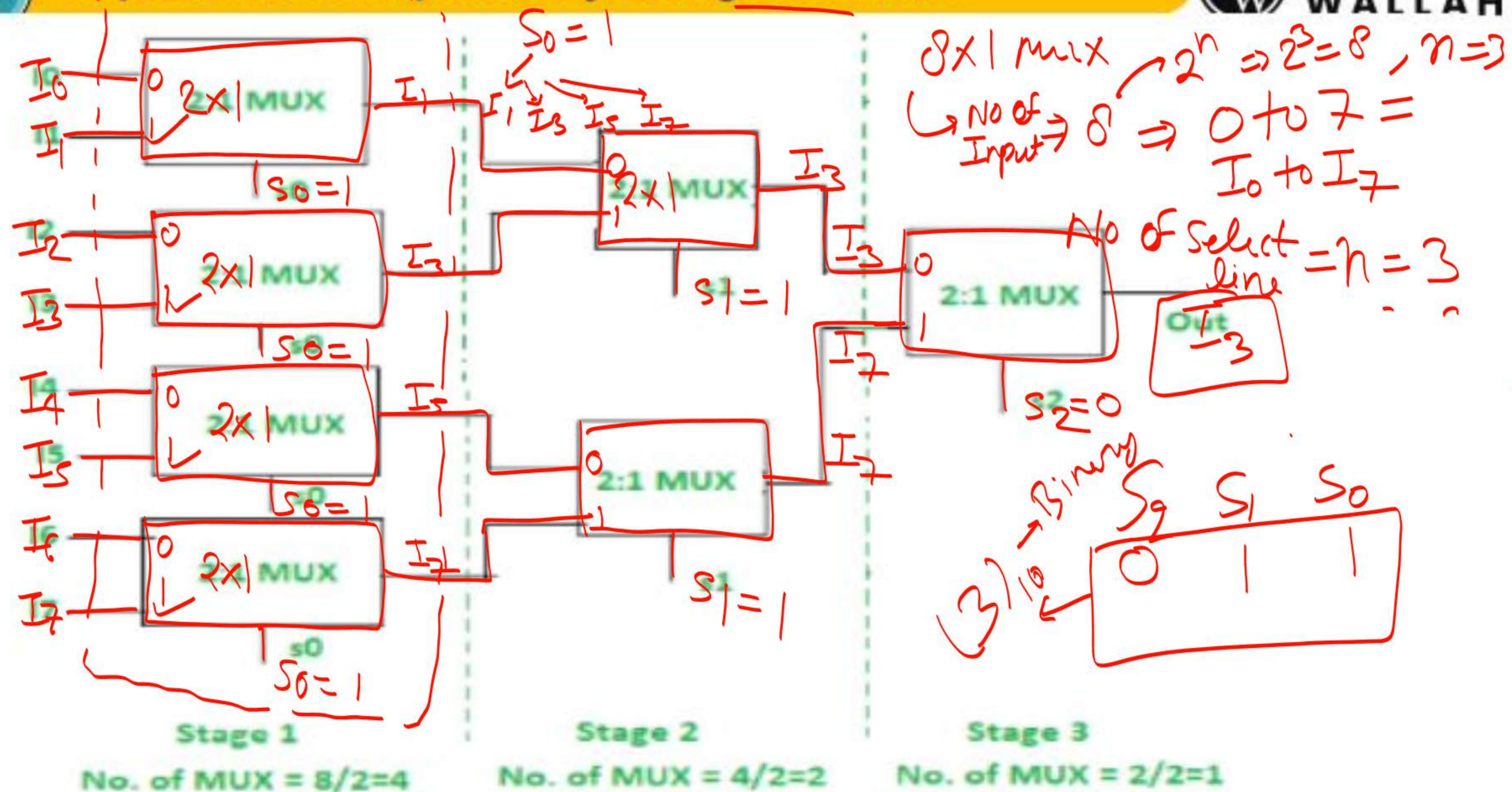




Topic : 8x1 Multiplexer by using 2 x1 mux



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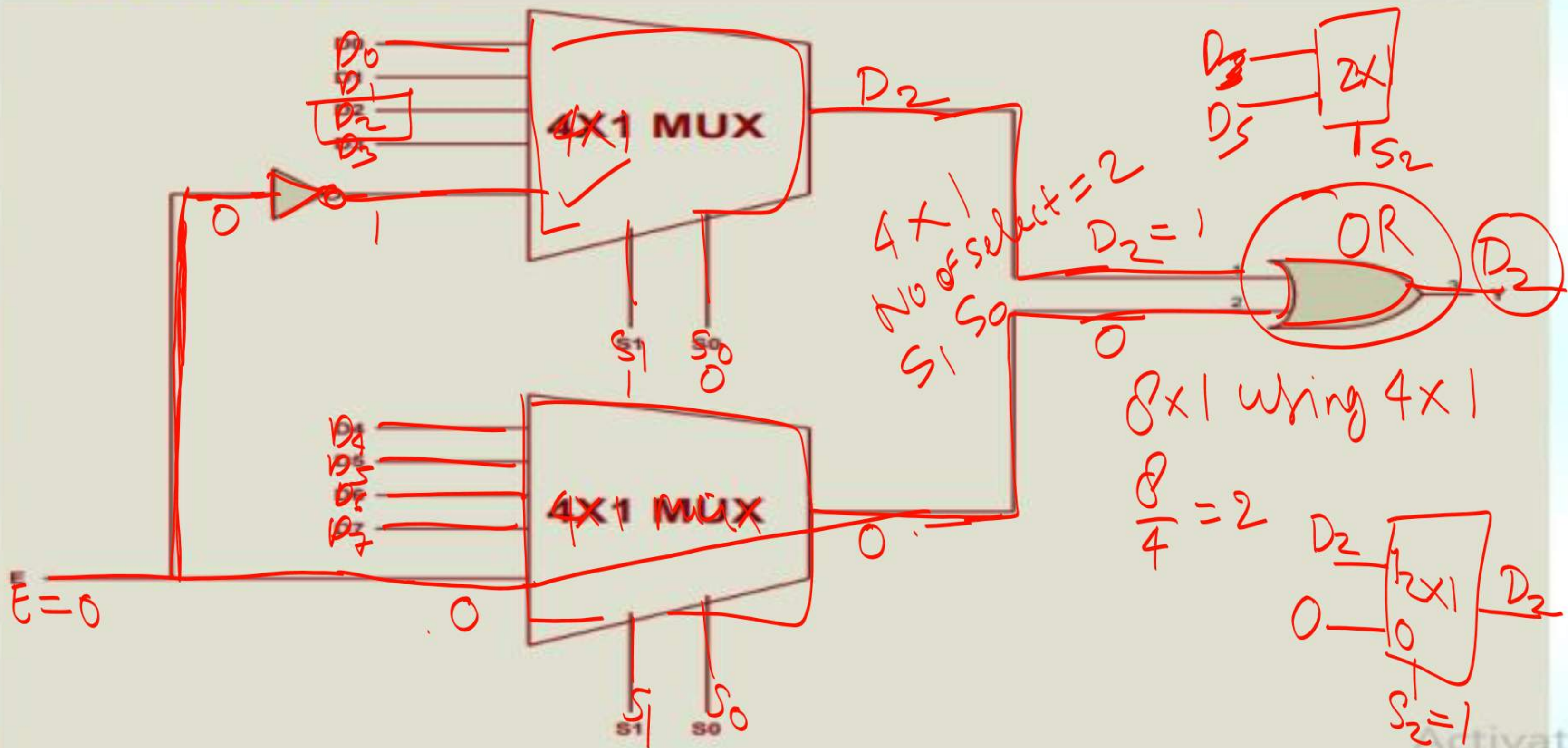




Topic : 8x1 Multiplexer by using 4x1 Mux- special case (Enable System)



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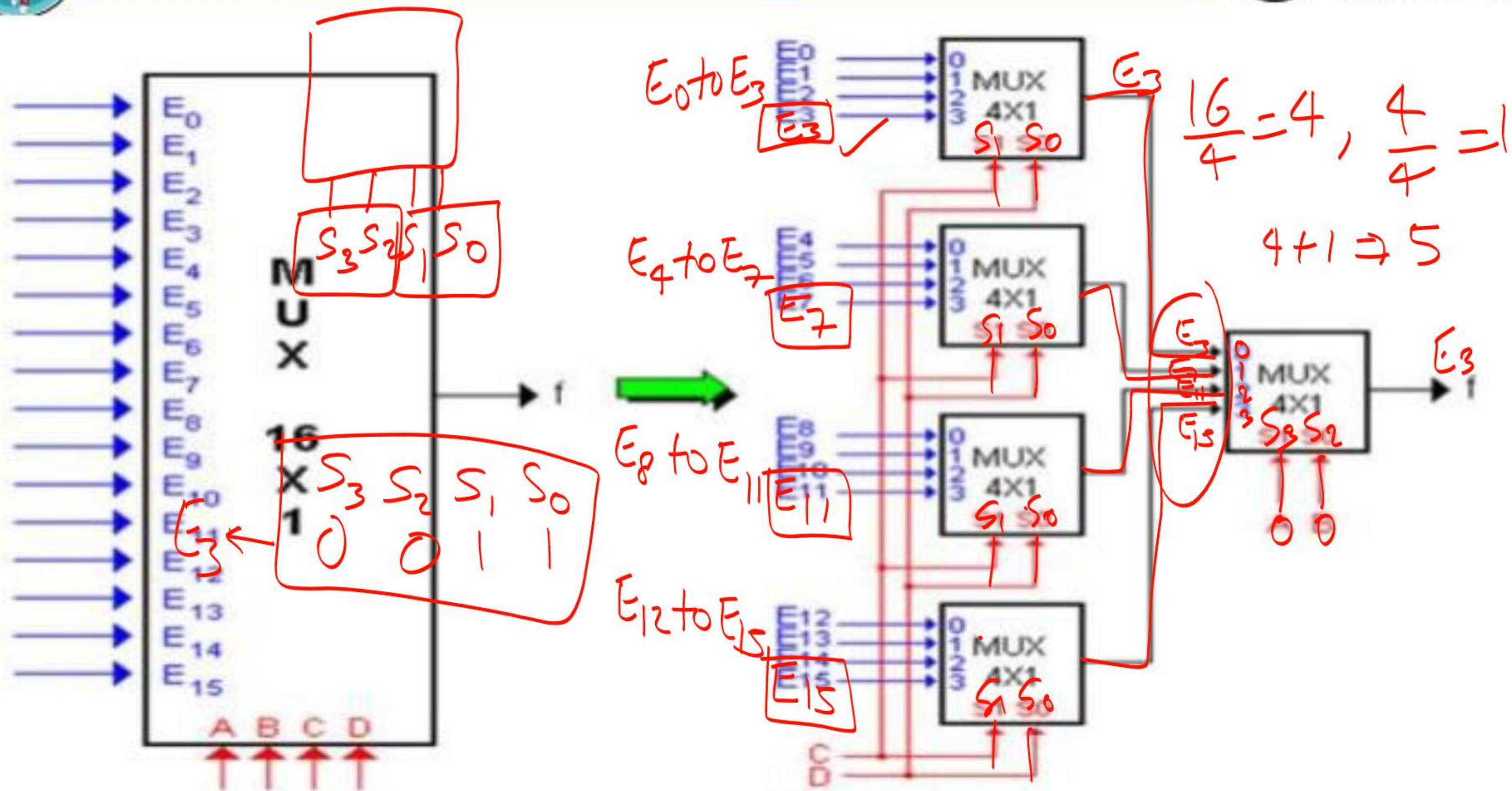




Topic : 16x1 Multiplexer by using 4x1 Mux



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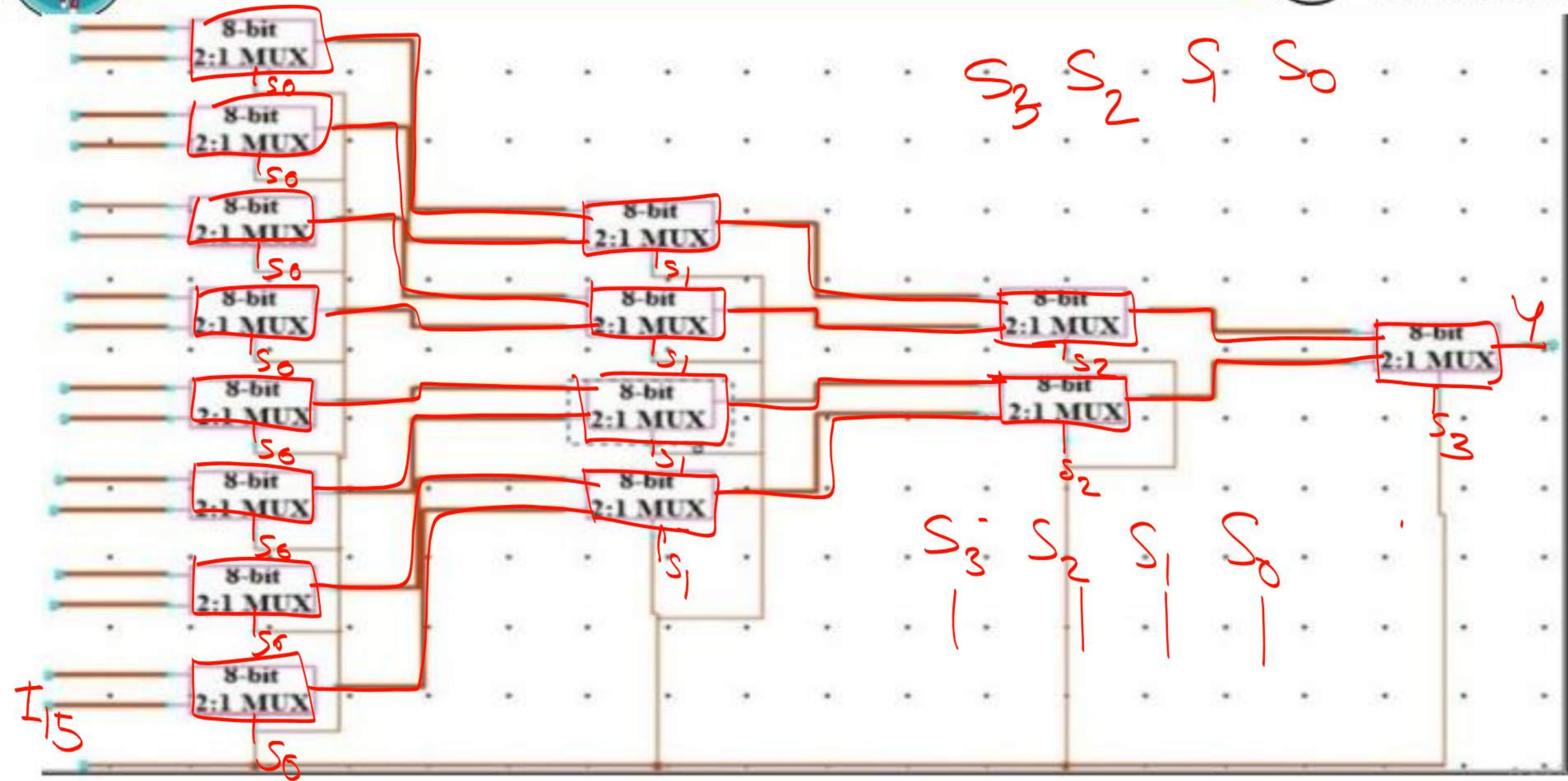




Topic : 16x1 Multiplexer by using 2x1 Mux



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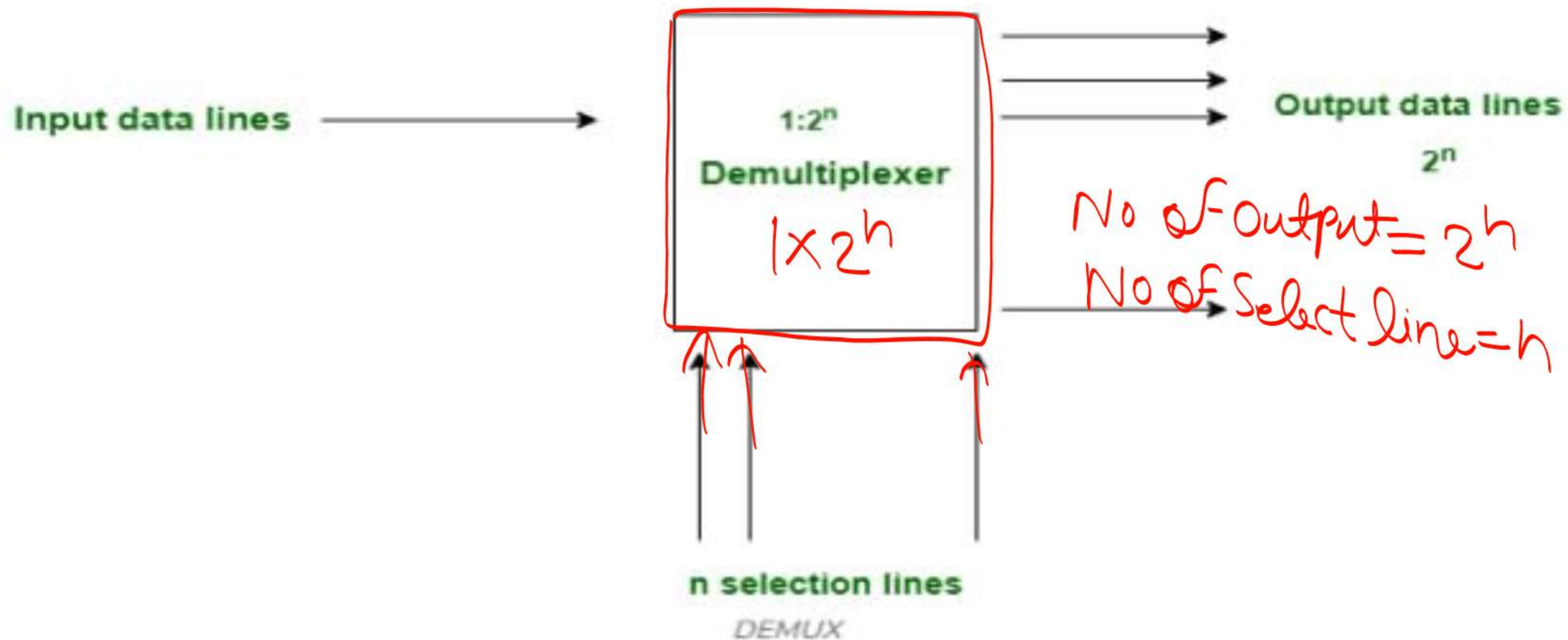
What is a DEMUX?

The DEMUX is a digital information processor. It takes input from one source and also converts the data to transmit towards various sources. The demultiplexer has one data input line. The demultiplexer has several control lines (also known as select lines). These lines determine to which output the input data should be sent. The number of control lines determines the number of output lines.



General Block Diagram Of A DEMUX

Here is the basic block diagram of a DEMUX as mentioned below.



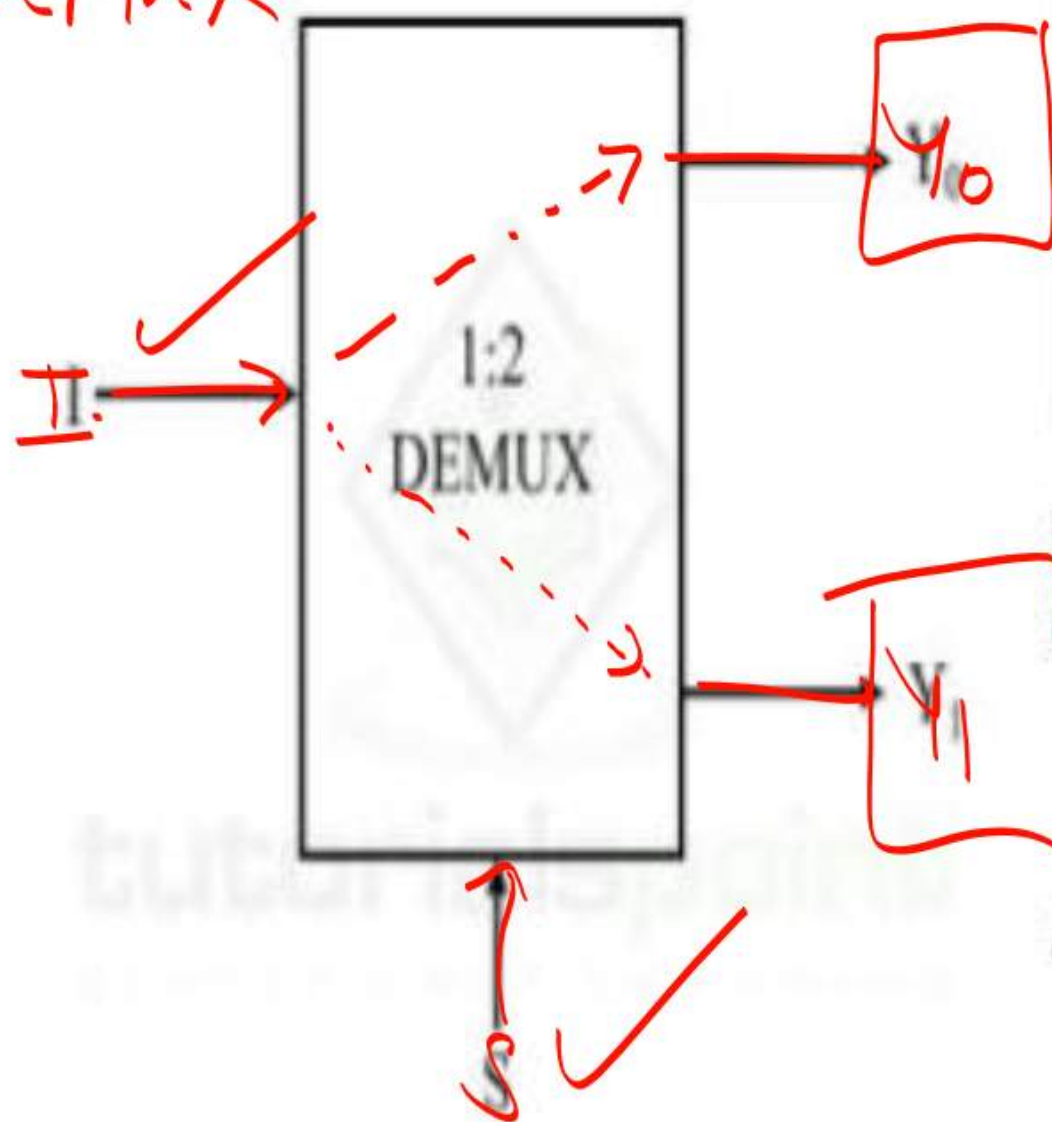


1x2 Demultiplexer

The operation of the 1x2 Demultiplexer can be analyzed with the help of its function table given below.

The functional block diagram of a 1x2 Demultiplexer is shown in

1x2 Demux



Select Line	Outputs	
S	Y_1	Y_0
0	0	I
1	I	0

From this function table of 1x2 Demultiplexer, we can directly derive the Boolean expression for each output as follow.

And,

$$S = 1 \\ \bar{S} = 0$$

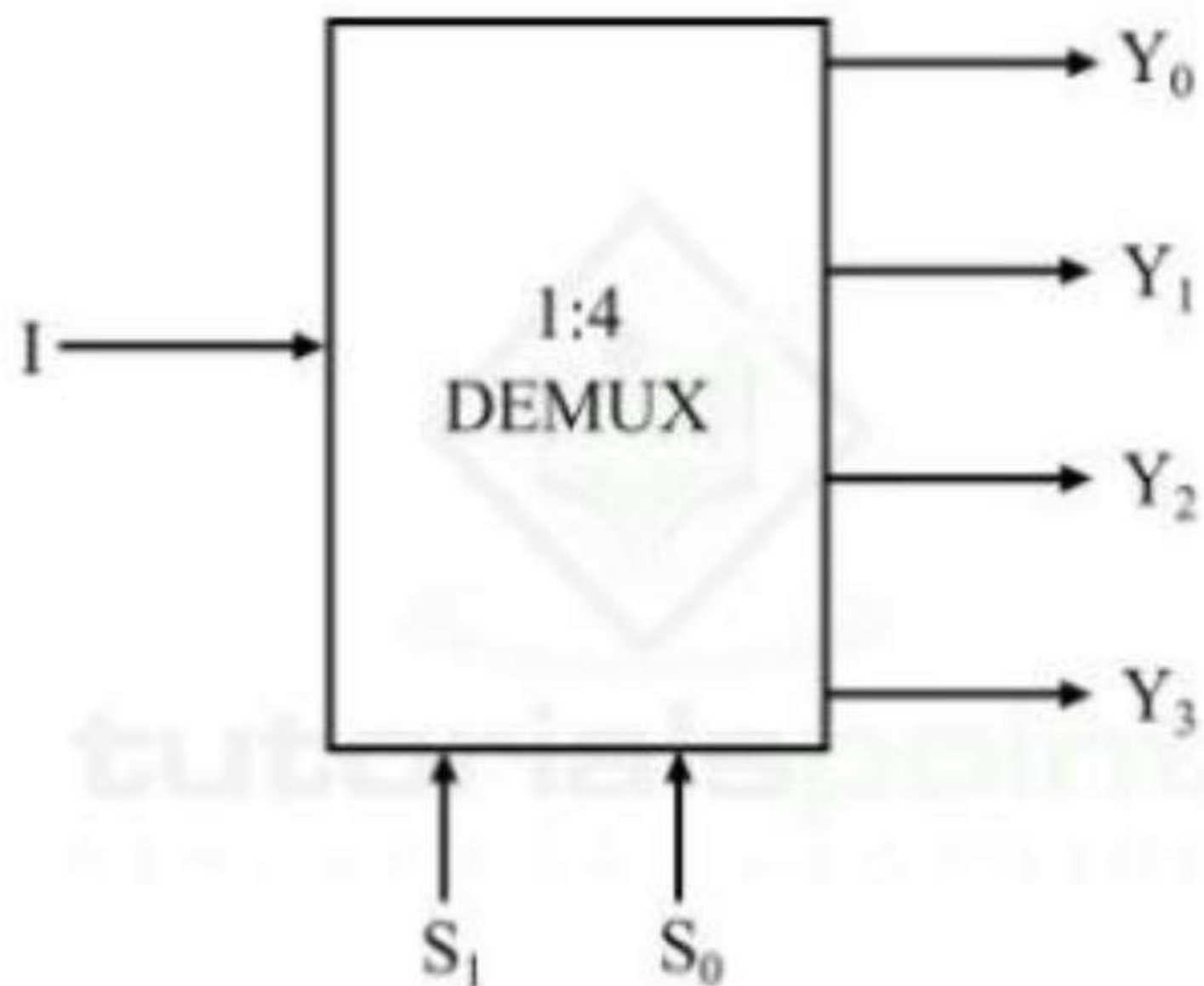
$$Y_0 = \bar{S}I = \bar{S} \cdot I$$

$$Y_1 = SI = S \cdot I$$



1x4 Demultiplexer

The functional block diagram of 1x4 Demultiplexer is shown in Figure-3.





The operation of the 1×4 Demultiplexer can be understood with the help of its function table given below.

Select Line		Outputs			
S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

From this truth table of 1×4 Demultiplexer, we can directly write the Boolean expression for each output as follow.

$$Y_0 = \bar{S}_1 \bar{S}_0 I$$

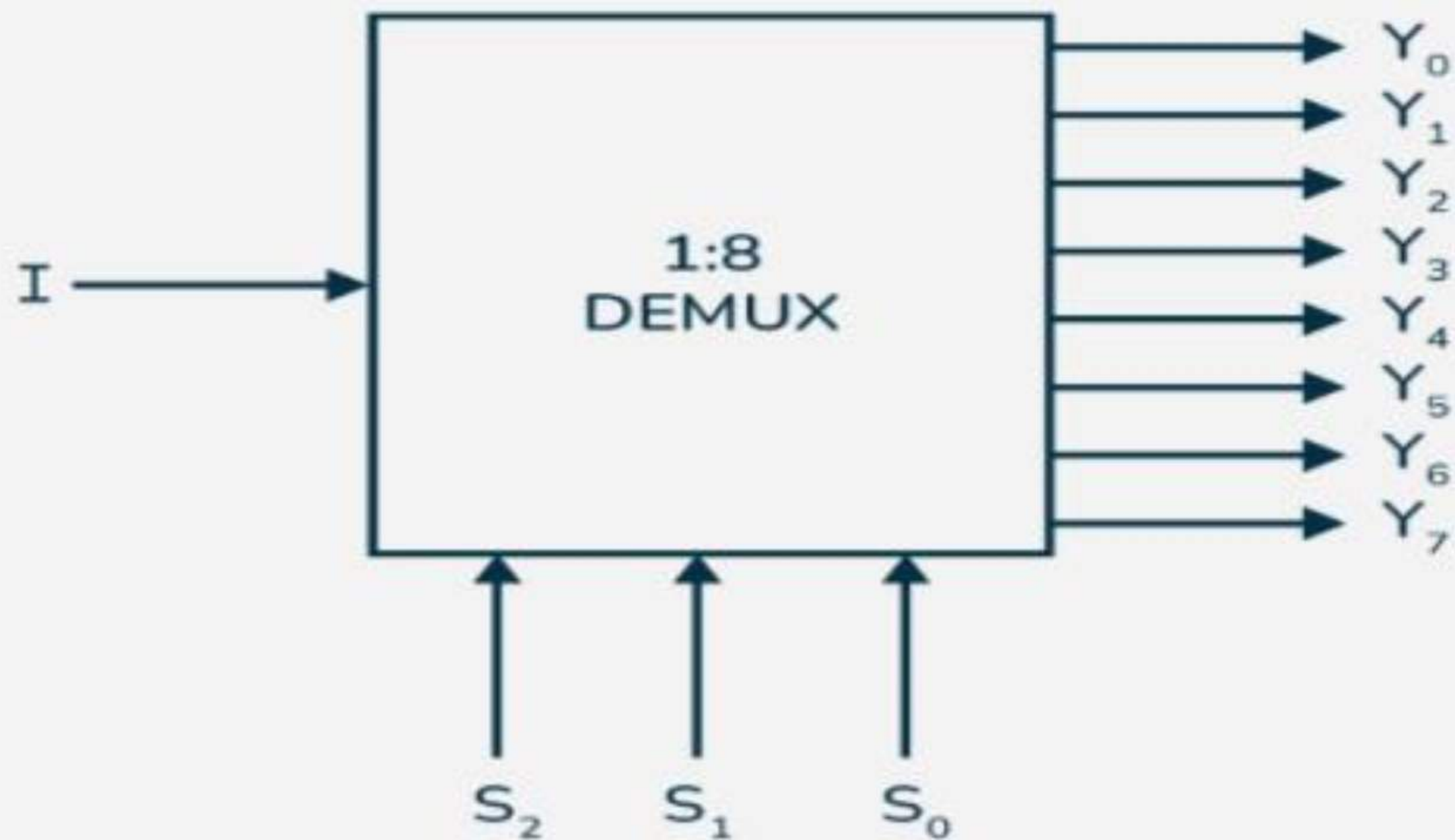
$$Y_1 = \bar{S}_1 S_0 I$$

$$Y_2 = S_1 \bar{S}_0 I$$

$$Y_3 = S_1 S_0 I$$



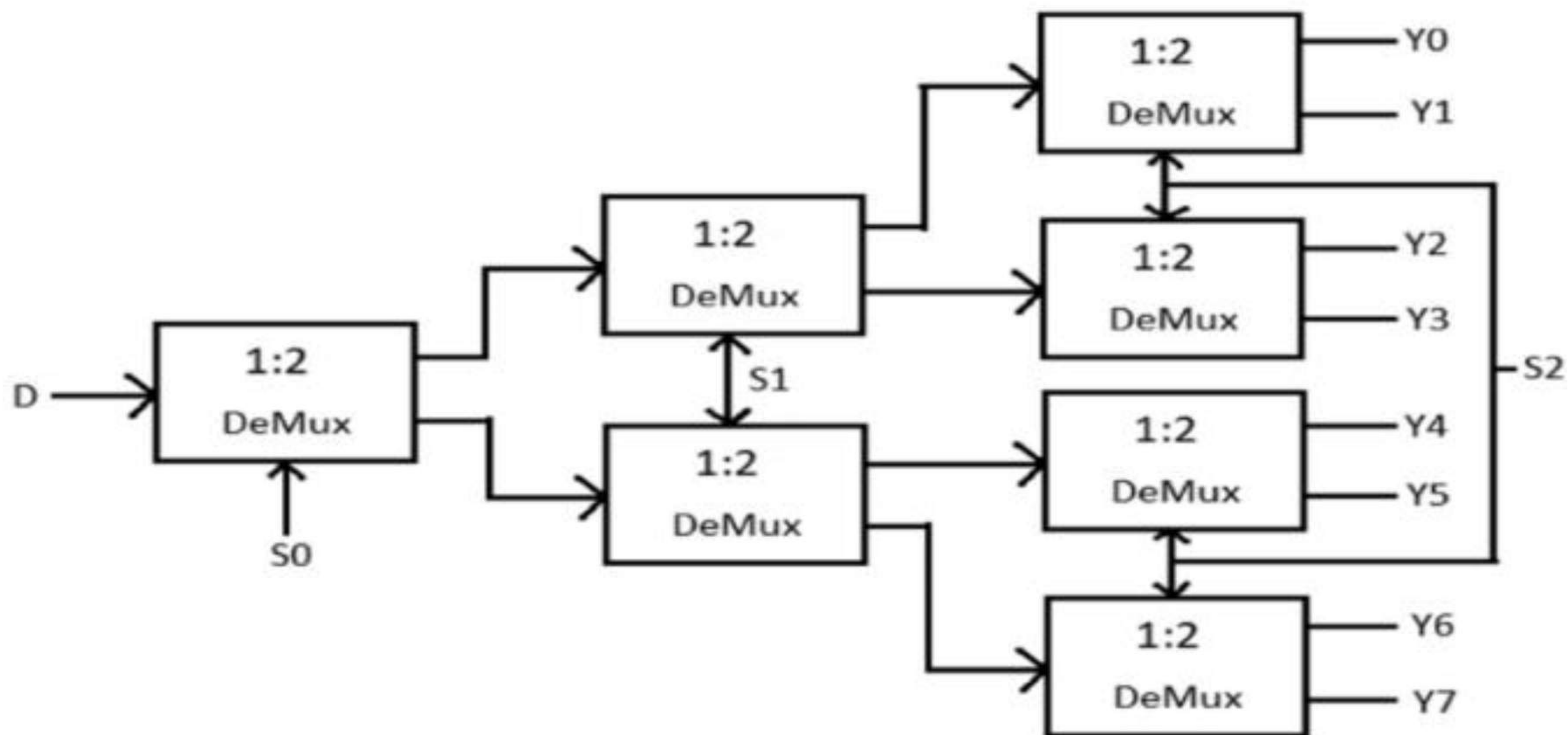
1:8 Demultiplexer

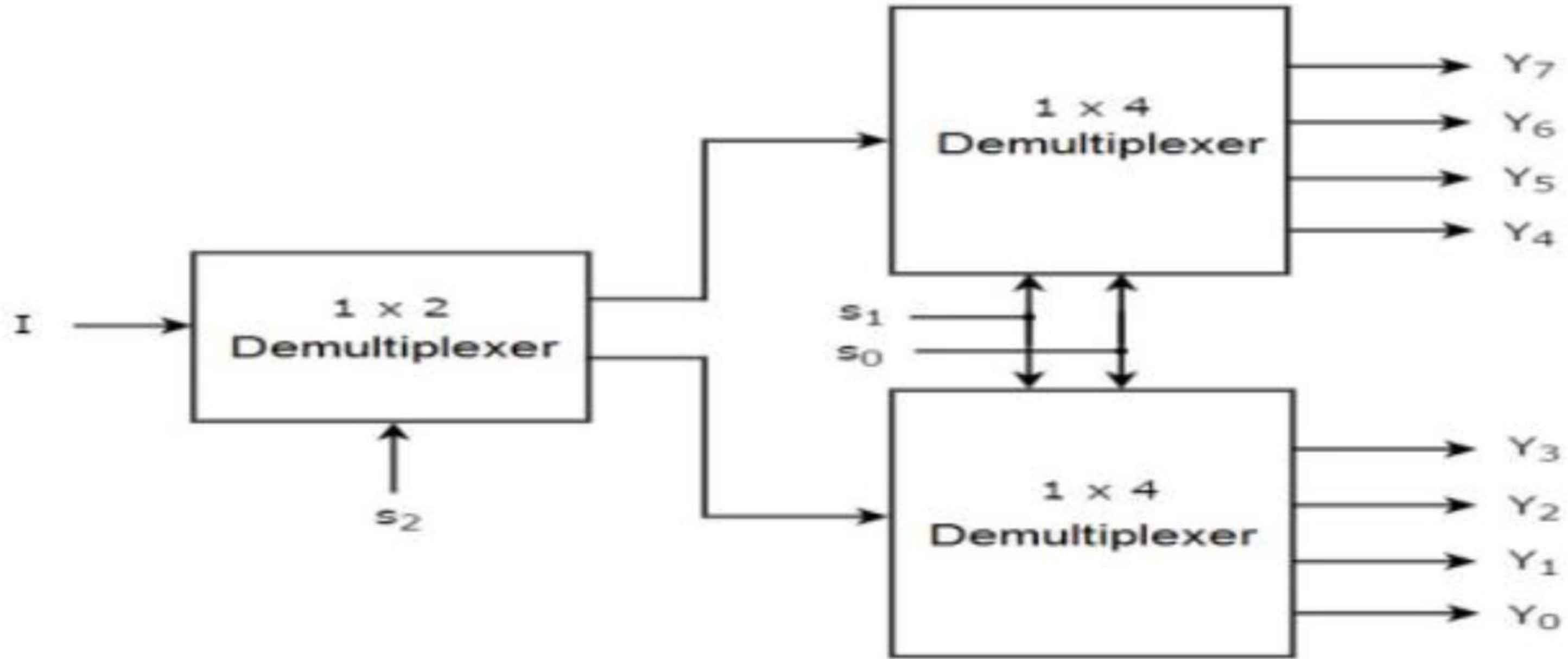




Let the 1×8 Demultiplexer has one input I , three selection lines s_2 , s_1 & s_0 and outputs Y_7 to Y_0 . The **Truth table** of 1×8 Demultiplexer is shown below.

Selection Inputs			Outputs							
s_2	s_1	s_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0





$8/4=2$, $2/4= 0.5$ That is why 1x8 does not create by only 1x4 and it will take help by 1x2 demux as well.



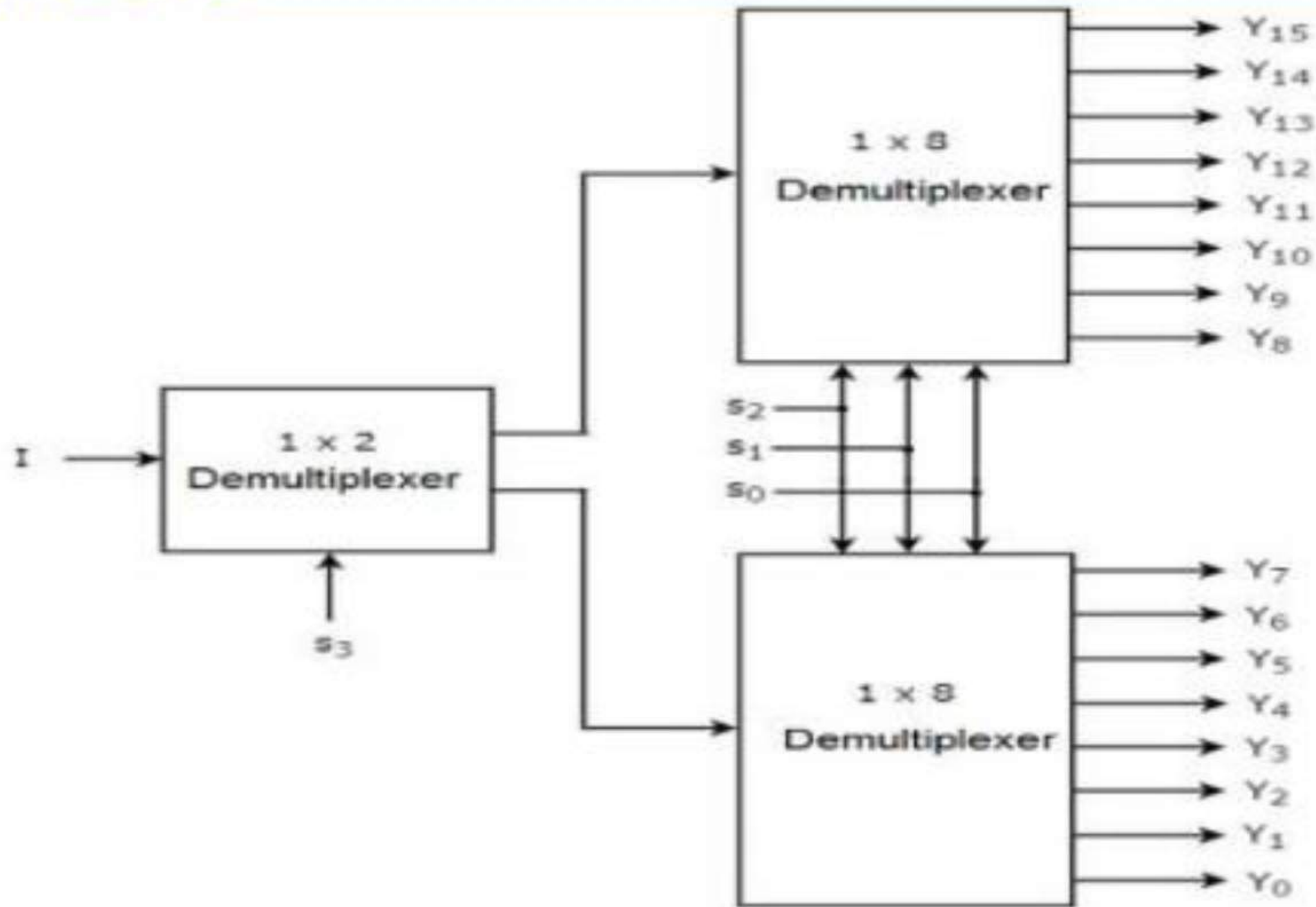
Create it own by you, It is Home-Work



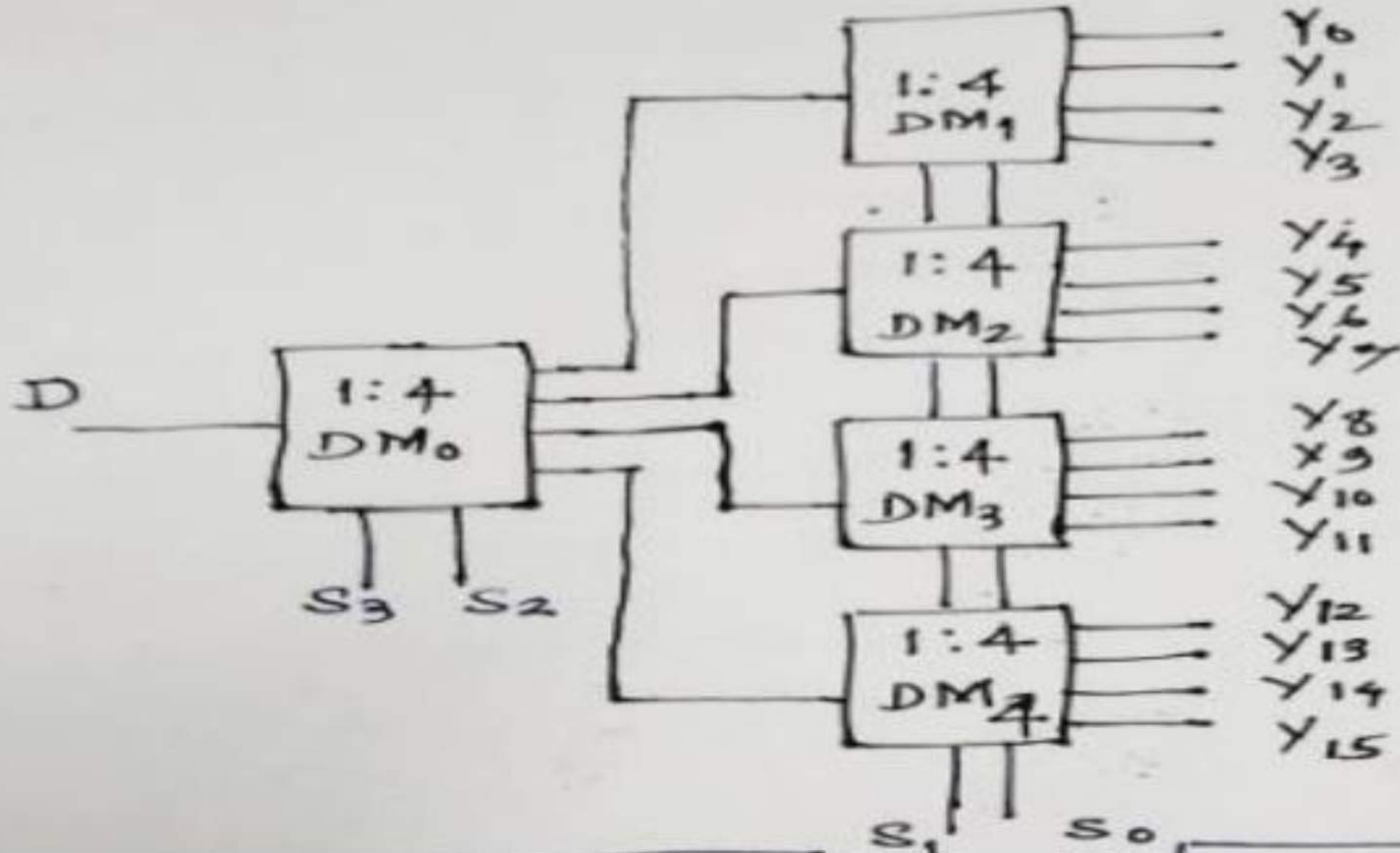
Topic : 1x16 Dmux using 1x8 Demux and 1x2 Demux- Special Case



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$16/8=2$, $2/16= 0.125$ That is why 1x16 does not create by only 1x8 and it will take help by 1x2 demux as well.





Advantages of the DEMUX

- The DEMUX increases the efficiency of the particular communication system as it takes data from a specific input source and distributes it to different sources.
- The DEMUX helps to separate the different signals from the mixed data sources. Then it distributes these data to different sources.
- DEMUX can decode the signal outputs of the [multiplexer](#), as the system works in a reverse way of the MUX.

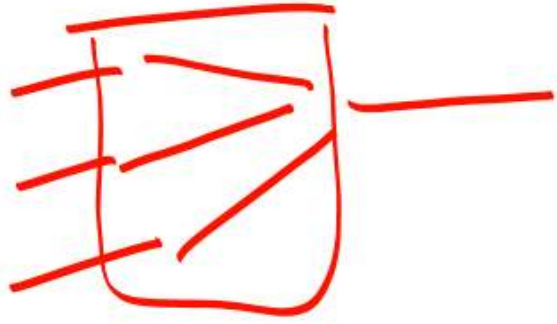
Disadvantages of the DEMUX

- The DEMUX may cause a wastage of bandwidth as it distributes the refined data in different channels. These channels can overlap with each other which leads to the loss of signal.
- The DEMUX may cause problems in the synchronization of signals. The data channels can overlap with each other which leads to the delay in the whole process.



Applications of DEMUX

- DEMUXs are majorly used to design the ALU circuits and parallel data segments.
- The DEMUXs convert the output of the MUX into the actual input. On the receiver end, the DEMUX can be used to verify the original form of the data to carry out the entire communication process.
- DEMUX helps to save the output. The output generally saved to the ALU. The output will be saved in the registers and the various storage units of the system.
- Each DEMUX has a connection with multiple registers which helps to store the processed data into it.
- The counterpart of the DEMUX regulates the data signal at the output stage of the DEMUX operation. These data can be retrieved later to read out parallelly.
- In the audio or video system, the DEMUX distributes them in different channels. In broadcasting, the DEMUX separates the composite signal into an individual one.



THANK YOU